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SMARTWATCH PCB PROJECT

Project Goal

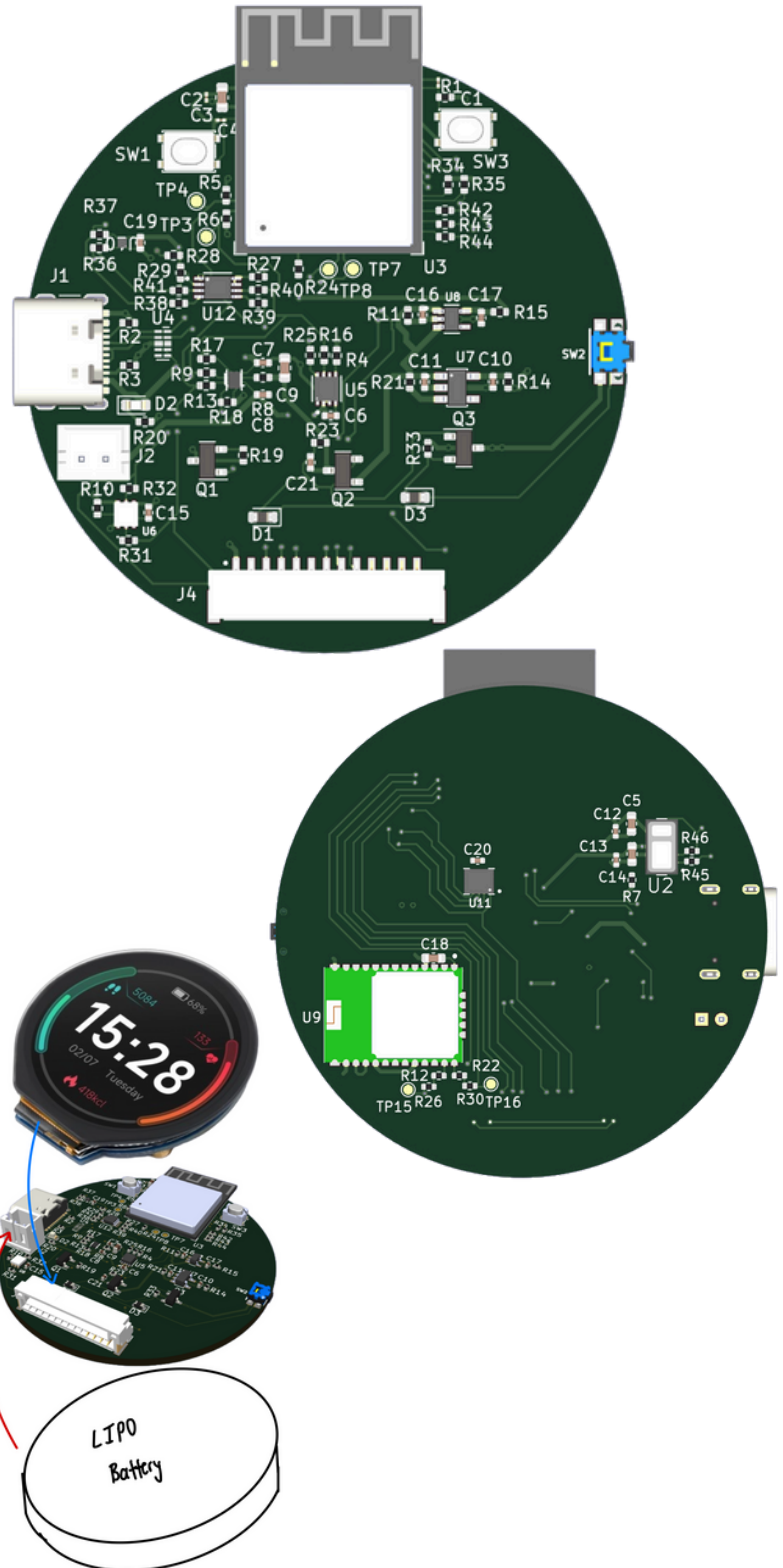
- Design and develop a compact, low-power smartwatch integrating multiple sensors and wireless connectivity for health monitoring and activity tracking, with efficient power management and implemented with an 1.28" circular LCD

Key Skills

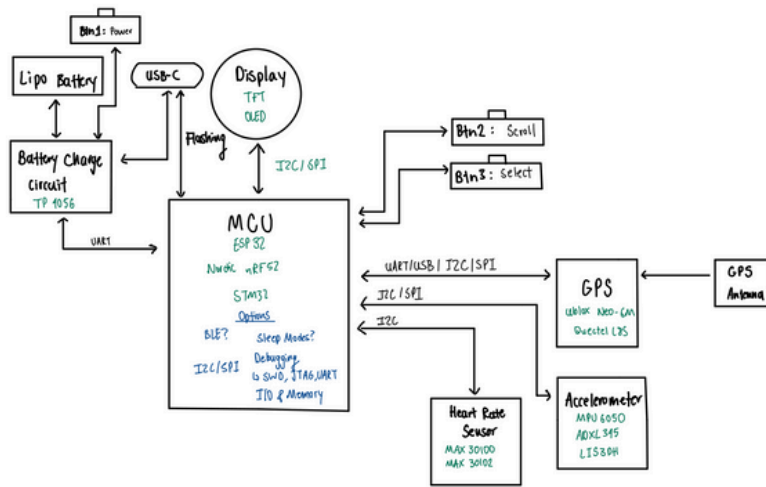
- PCB Design (KiCAD):** 4-layer board layout in KiCAD with signal integrity, power distribution, and compact routing
- Sensor Integration:** I²C and SPI communication with IMU, heart rate/SpO₂ sensor, GPS, compass, and ambient light sensors
- Power Management:** USB-C charging, soft-latch power control, battery fuel gauging, and multiple voltage rails (1.8 V, 3.3 V)
- Hardware Validation:** Bus topology design, level shifting, testpoint planning, and interface debugging
- Protocols & Interfaces:** USB, I²C, SPI, UART, BLE, Wi-Fi, GNSS

Project Features

- Designed a 1.28" circular 4-layer PCB integrating the **ESP32-S3-MINI-1** with Wi-Fi, BLE, and USB support
- Integrated sensors via I²C/SPI for heart rate (MAX30102), IMU (LSM6DSOX), magnetometer (AK0991x), GPS (MAX-M10S / CAM-M8Q), and ambient light monitoring, with appropriate **level shifting** for 1.8 V devices
- Implemented **USB-C** power input with battery charging, **power path** management, fuel gauge, and **soft-latch** power control circuit, LDOs for regulated 3.3 V and 1.8 V power rails
- Connected a 1.28" SPI TFT LCD with capacitive touch interface for user interaction
- Focused on signal integrity, proper **test point** and **jumper** placement, and **controlled impedance** routing for high-speed and RF lines



Initial System Diagram



I2C Pull-Up Resistor Calculation/Bus Design

IC	Addr	Vol	VOH	t _{rise}
3.3V	VEML6030	0x10	0.4	1-3
	MAX17046	0x34 (7bit)	0.5	1-4
	0x26198	0x68 (7bit)	0.4	1-3
	0x04 (7bit)			1000 ns
	GC9A01 (10bit)	0x15 (7bit)	0.6	1-4
				1000 ns
1.8V	AK09418	0x0C (7bit)	0.3	-
	MAX30102	0x57 (7bit)	0.4	-
				1000 ns

$I_{OL} = 3mA$

3.3V: $R_{pullup\ min} = \frac{V_{CC} - V_{OL(max)}}{I_{OL}} = \frac{3.3V - 0.4}{0.003} = 900\ \Omega$

1.8V: $R_{pullup\ min} = \frac{1.8V - 0.4}{0.003} = 467\ \Omega$

$C_b = 15pF \times 4\ devices + 20pF + 20pF$

3.3V: $R_{pullup\ max} = \frac{t_{rise}}{0.8473 \times C_b} = \frac{1\ \mu s}{0.8473 \times 100pF} = 29.8K$

$R_{pullup\ max} = 11.8K$

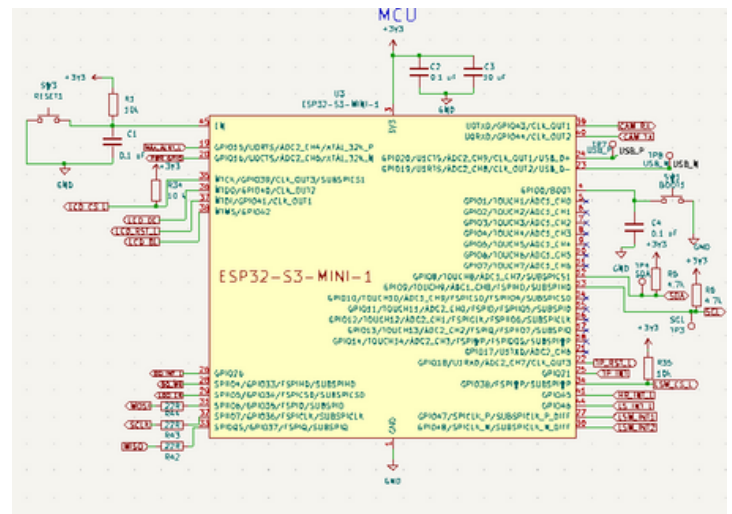
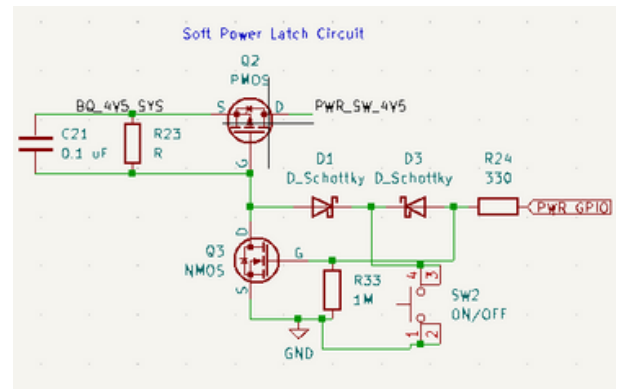
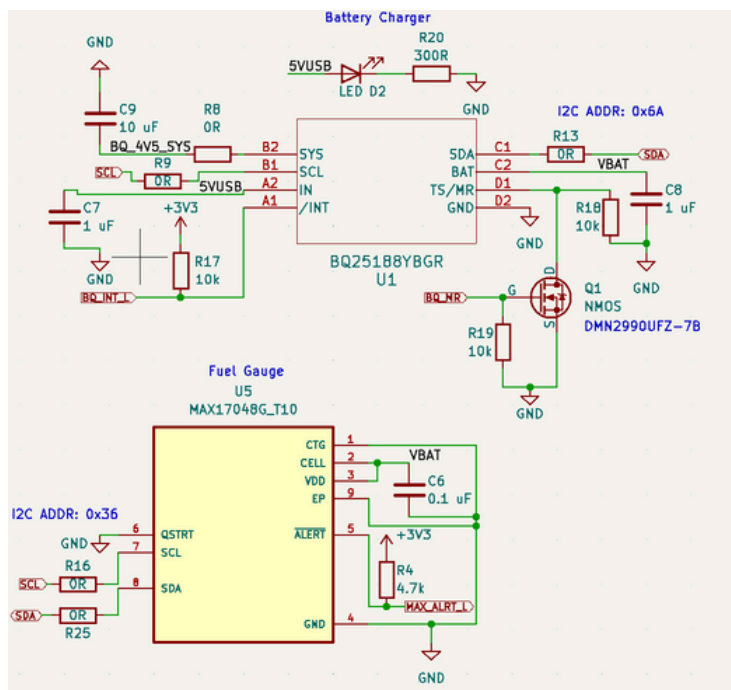
3V3 LDO Junction Temperature Calculation

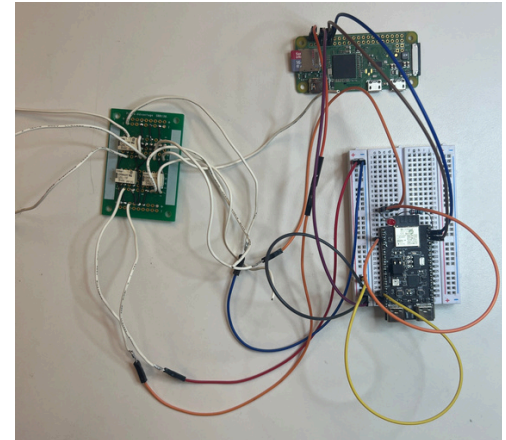
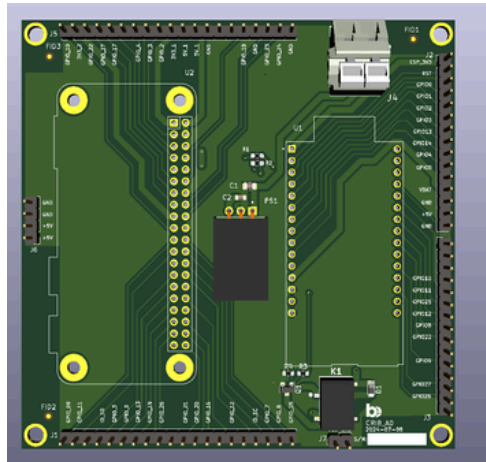
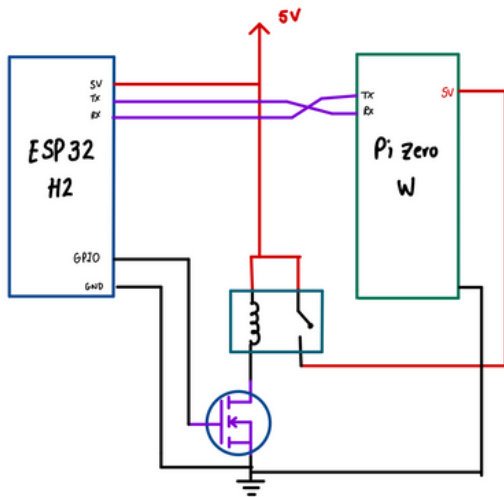
$$\begin{aligned}
 V_{in} &= 4.5V \\
 V_{out} &= 3.3V \\
 I_{load} &= 0.6A \\
 I_{GND} &= 8 \times 10^{-6}A \\
 P_{Dmax} &= (4.5V - 3.3V) \times 0.6A + 4.5V \times 8 \times 10^{-6}A \\
 &= 0.72W + 3.6 \times 10^{-4} \\
 P_{Dmax} &= 0.72036W \\
 \text{Thermal Resistance (Junction to case)} &: 96^{\circ}C/W \\
 \text{Junction temp} &= 0.72036W \times 96^{\circ}C/W \\
 &= 69.15^{\circ}C + 25C \\
 &= 94.15^{\circ}C \checkmark
 \end{aligned}$$

1V8 LDO Junction Temperature Calculation

$$\begin{aligned}
 V_{in} &= 4.5V \\
 V_{out} &= 1.8V \\
 I_{load} &= 0.0012A \\
 I_{GND} &= 3.9 \times 10^{-6}A \\
 P_{Dmax} &= (4.5V - 1.8V) \times 0.0012A + 4.5V \times 3.9 \times 10^{-6}A \\
 &= 3.4155 \times 10^{-3}W \\
 \text{Junction thermal resistance} &: 256.5^{\circ}C/W \\
 \text{Junction temp} &= 256.5^{\circ}C/W \times 3.4155 \times 10^{-3}W + 25C \\
 &= 0.876C + 25C \\
 &= 25.876C \checkmark
 \end{aligned}$$

[VIEW SCHEMATIC PDF](#)





Project Goal

- Reduce the sleep current of the Pi Zero W from current 180 mA for battery powered applications
- Design for future expansion such as adding sensors, support for other MCUs
- Track time spent asleep during operation

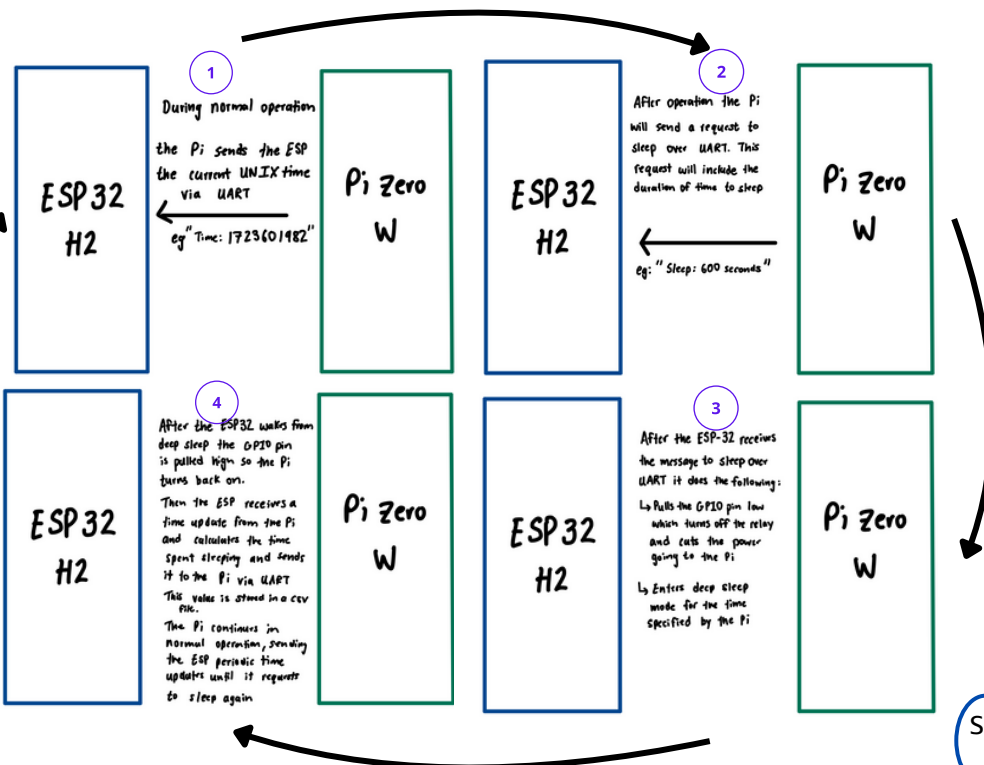
Key Skills

- PCB Design (KiCAD)
- ESP32
- UART (Software & Elec)
- C Programming
- FreeRTOS
- RTC
- Raspberry Pi
- Python

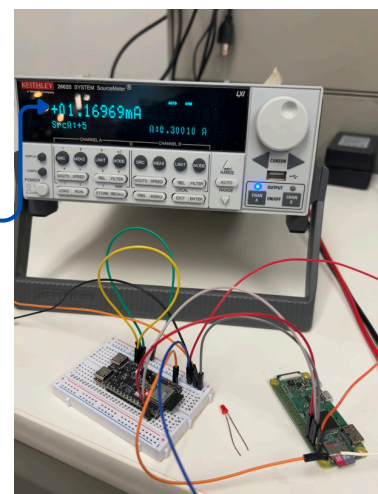
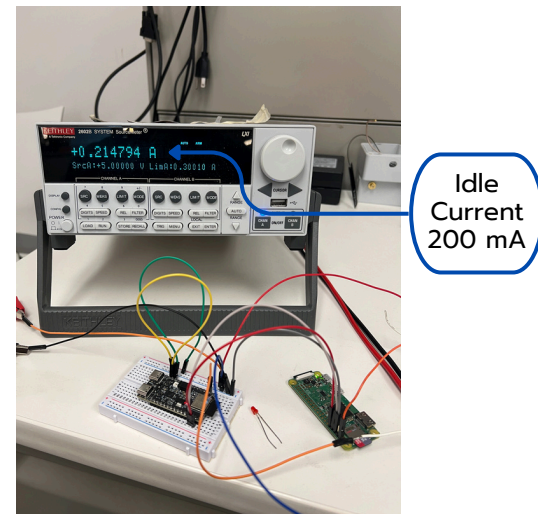
Project Features

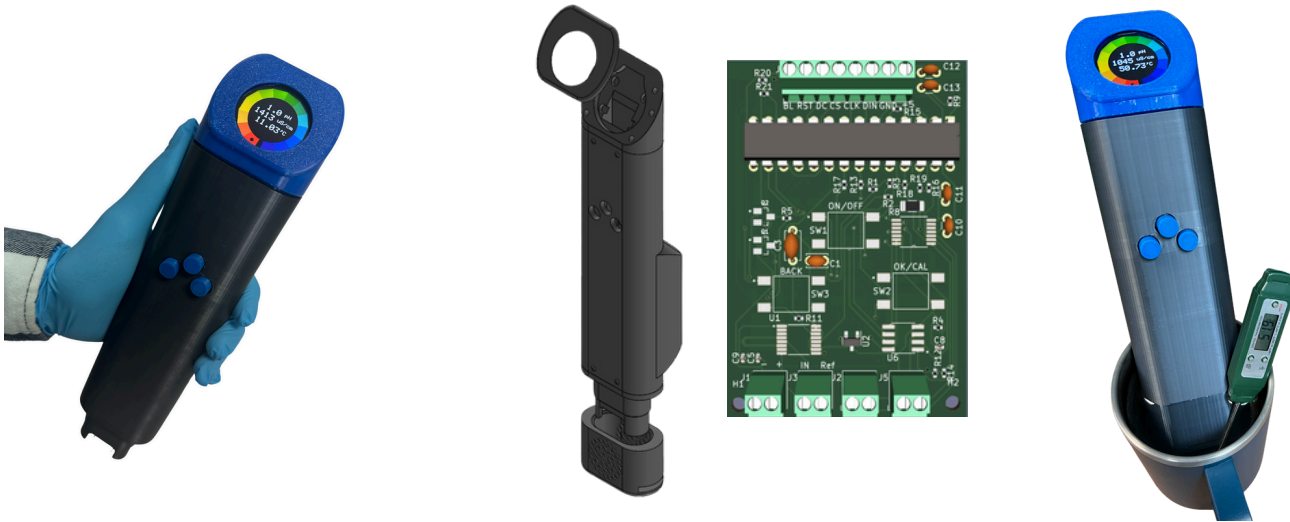
- **UART** interface between ESP and Pi Zero
- Pi power controlled by Signal Relay
- Relay controlled by **MOSFET** and ESP GPIO
- On board 24V to 5V DC Converter
- Breakout Headers for expansion
- V2 PCB features support for Adafruit M0
- Sleep Current of **1 mA**

Sequence of Operation



Idle vs Deep Sleep Current





Project Goal

- Design a handheld device capable of measuring the parameters of a water sample such as PH, Temperature, Conductivity and ORP
- Modular design such that it can be interchanged for different applications and parameters

Key Skills

- Low-Voltage PCB Design (KiCAD)
- High Resolution Sensor Measurement (ADC)
- Signal Amplification (Op-Amp)
- Digital Circuits (Latch Circuit)
- SPI (LCD)
- Fusion360
- C++
- ATmega328

Project Features

- 2 Layer PCB to interface sensors with MCU and LCD
- 12 Bit 4 channel **ADC** for increased resolution
- Power latch circuit for auto-shutoff
- Op-amp circuit for PH Sensor amplification
- Custom GUI to display measurements
- 3D printed enclosure for prototype

Version 1



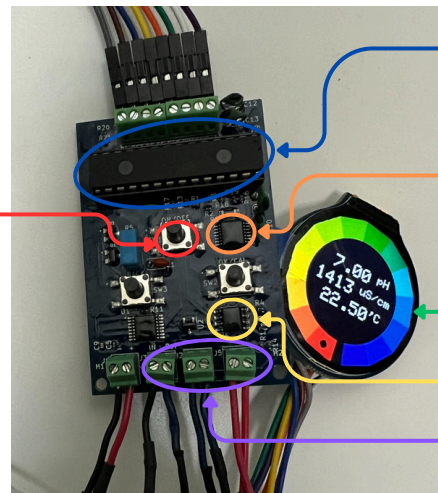
Increased Resolution with ADC



Shrunk PCB size with improved routing and SMT components

Power Button (Latch)

Version 2



Microcontroller

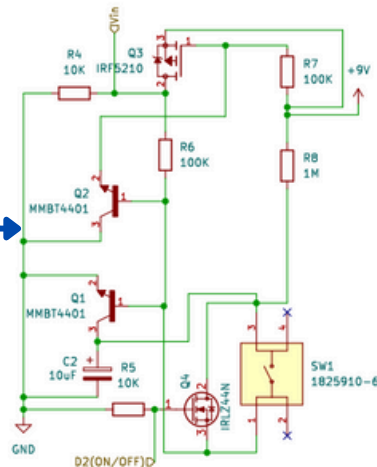
Analog-to-Digital Converter

LCD Display

Op-Amp

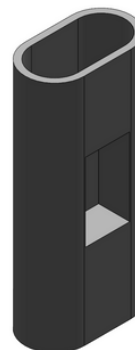
Sensors

Power Latch Circuit

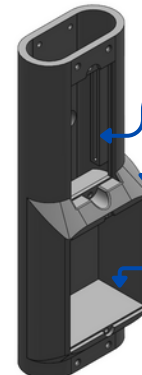


Designed Power Latching Circuit with power button that allowed for timed auto-shutoff of device, as a result saved battery usage

Version 1



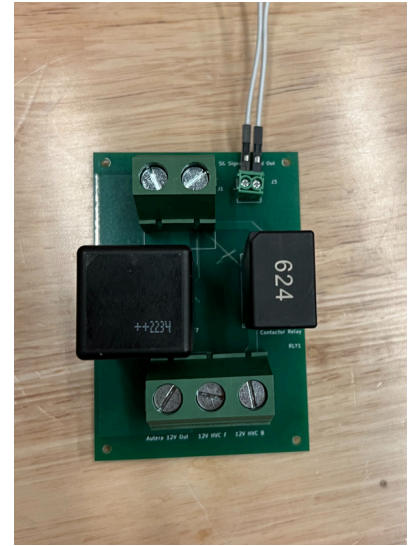
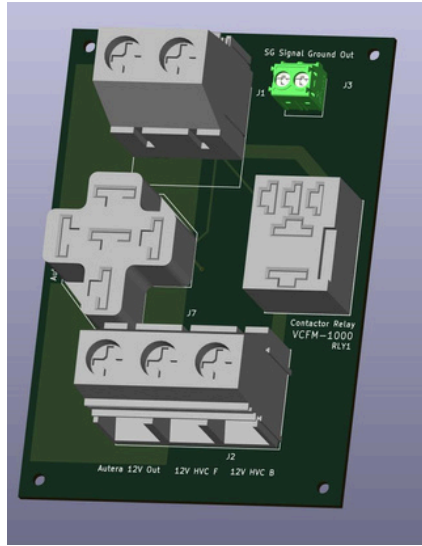
Version 2



Separate access to PCB for maintenance

Ergonomic and better fit in hand

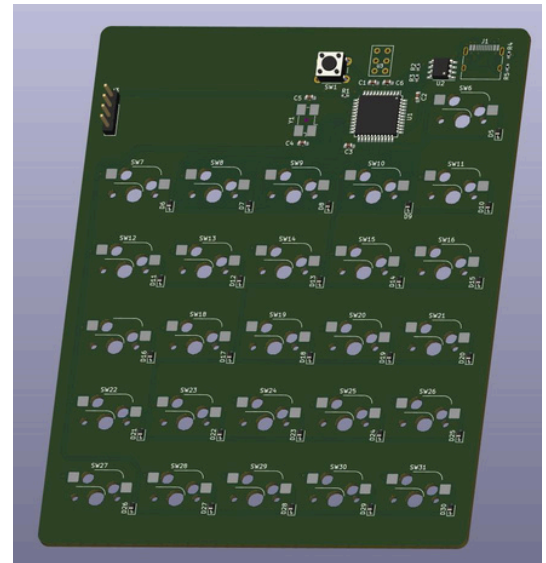
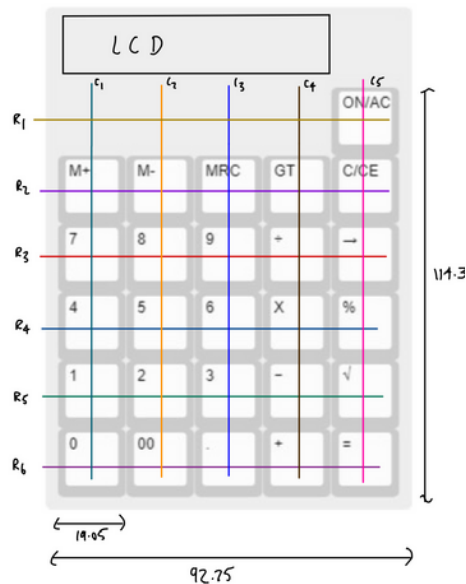
Changed opening from 9V battery to 6xAA for increased current capacity



Project Features

- Automotive grade **Relays** to control Power going to Autera and Contactors
- Large terminal blocks for 10 AWG wire
- **nMOS** to control relays from Speedgoat input

BRAILLE KEYBOARD



Project Features

- **ATmega32U4** for detecting input from keys and performing arithmetic
- **Diode** array for preventing key ghosting
- **ISP** header as alternate flashing method
- **USB** connection for power and serial communication with computer
- **I2C** LCD for debugging purposes